Lab Assets (Runnable Classroom Kits)

Notes teach; **these folders run**. Start every major concept with a **customer ticket**, then `starter/` or `broken/`.

Layout convention

```
15_Lab_Assets/
  <nn>_<name>/
    README.md     # ticket (start here)
    starter/      # incomplete — learners extend
    broken/       # intentional bugs — debug lab
    solution/     # reference answers (do not share; pack script excludes)
    seed/         # sample CSV/JSON/SQL when needed
```

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Kit	Sessions	Ticket	Status
01_terminal_workspace	~01–02	Messy Desktop → Clean Workspace	**Ready**
02_hello_javac	~03–05	First `HelloJava` compile/run	**Ready**
03_canteen_calculator	~06–10	Campus Canteen Calculator	**Ready**
04_casting_bugs	~09	Marks Entry Keeps Crashing	**Ready**
05_atm_menu	~11–15	Mini ATM Menu	**Ready**
06_string_parser	~16–17	Registration Field Cleaner	**Ready**
07_marks_manager	~18–20	Student Marks Manager (arrays)	**Ready**
08_methods_refactor	~21–23	Refactor ATM Into Methods	**Ready**
09_bank_account_oop	~24–30	BankAccount (private balance)	**Ready**
10_employee_inheritance	~31–33	Staff hierarchy (`extends` / `super`)	**Ready**
11_campus_wallet	~34–36	Abstract `Account` + `Payable` + poly loop	**Ready**
12_collections_inventory	~37–40	Canteen inventory (`List` / `Set` / `Map`)	**Ready**
13_safe_input_exceptions	~41–42	Never Crash on Bad Input	**Ready**
14_expense_files	~43	Expense CSV append	**Ready**

Kit	Sessions	Ticket	Status
15_git_portfolio_hygiene	~44–45	Java/Maven `.gitignore` + README	**Ready**
16_jdbc_library_crud	~55–58	Library CRUD + no concat SQL	**Ready**
17_spring_hello_rest	~66–70	Two JSON endpoints + in-memory GET list	**Ready**
18_jpa_product_crud	~71–74	Product JPA catalog + DTO / `@Valid` / Postman v1	**Ready**
19_spring_auth_tasks	~75–80	JWT register/login + protect POST; Lab 06 freeze (400/404 + demo bundle); Lab 07 Actuator health + env / PaaS / M0 on the same API	**Ready**
`20_junit_utils`	~91–92	Tests Must Catch the Bug	Planned
`21_ai_review_stub`	optional after S95	Optional AI recommend endpoint	Planned

MySQL seed labs stay in `MYSQL/` (required via the FS bridge). Frontend kits: `FRONTEND/labs/` · `FRONTEND/REACT/labs/`.

Kits 20+ are created when that calendar phase is authored (teaching-order rule: session **S** kits may only use syntax from sessions **1…S**). Kit 19 core path is register / login (BCrypt + JWT), public GET `/products`, protected POST `/products`. Lab 06 freeze also requires GET by id 200/404, POST invalid with Bearer → 400, README/demo bundle, and peer break-auth on `broken/`. Lab 07 (same kit, later sessions) adds Actuator `/actuator/health`, host env, a PaaS attempt or blocked note, and capstone M0 — it does **not** change Lab 06 freeze Done when.

How to use a kit

1. Open the kit folder (or copy `starter/` / `broken/` into your workspace).
2. Read the kit `README.md` ticket first — code before long theory.
3. Work the ticket solo or in pairs.
4. Compare with `solution/` only after you finish (or when told in class).

Student pack policy

`solution/` folders are **excluded** by `scripts/pack_student_share.sh java` and should not be shared. Starter/broken/seed remain.

Map to modules

Lab kits live under this folder. Session order stays in Session_Calendar_100.md. Modules link kits from **Mini-Project** (and practice) — there is no separate Classroom Ticket section.